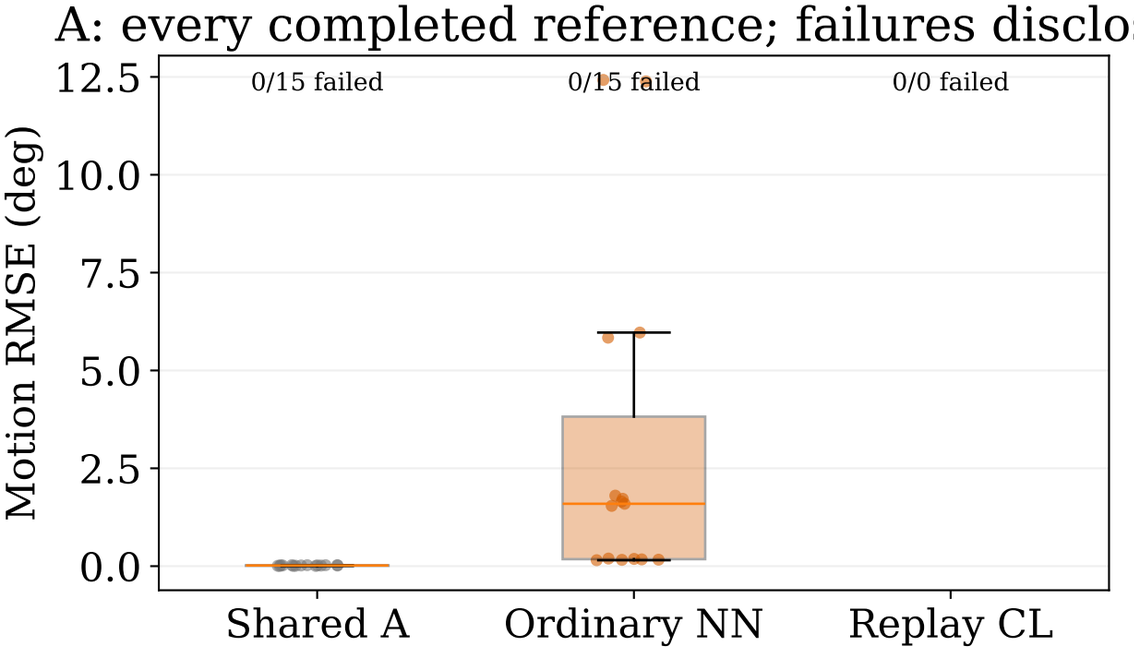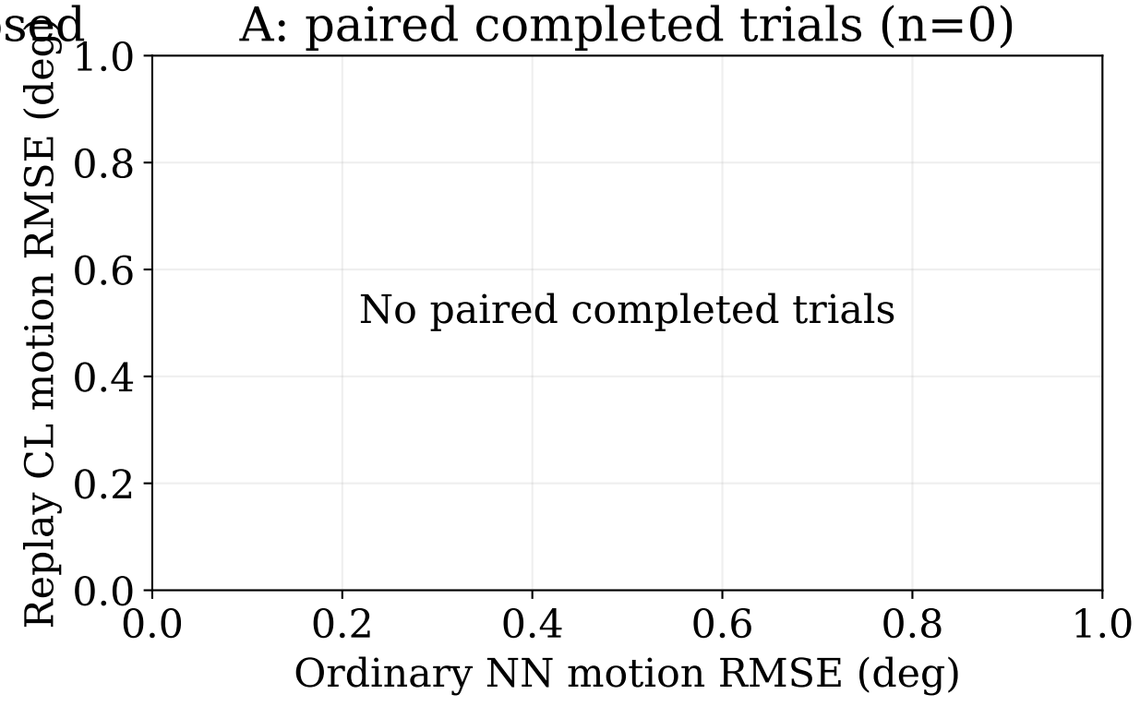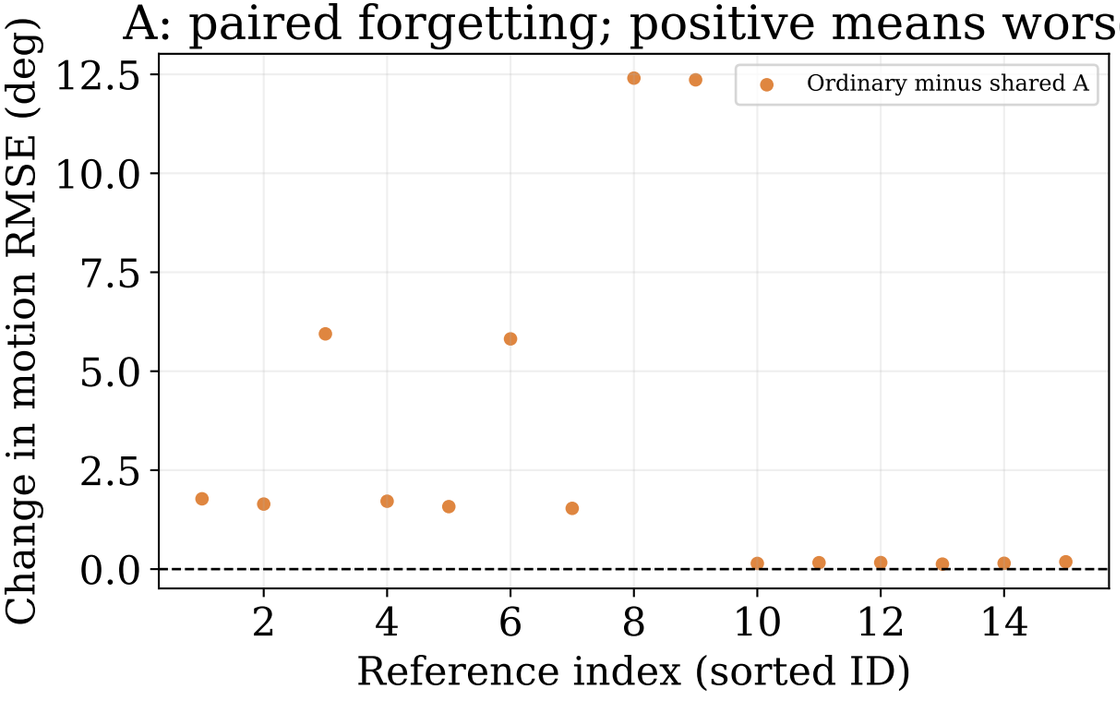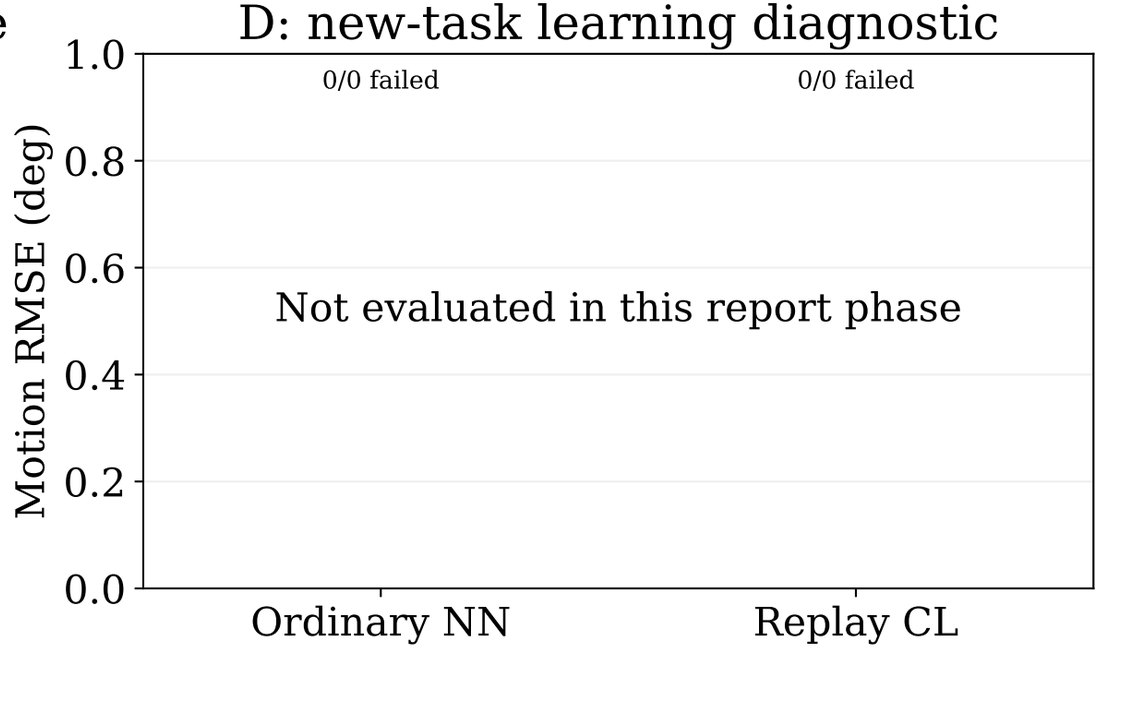VALIDATION ONLY — all trials retained; failed trials have no invented final RMSE